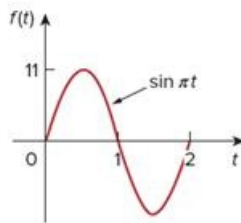


Problem

Find the Fourier transform of the “sine-wave pulse” shown in Fig. 18.36.

Figure 18.36



Step-by-step solution

Step 1 of 5

The given “sine-wave pulse” is,

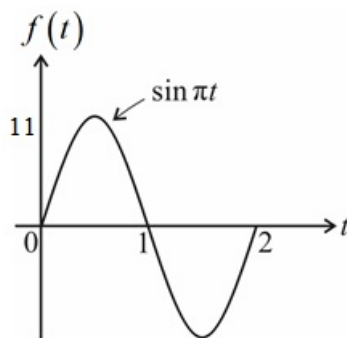


Figure 1

Step 2 of 5

The function $f(t)$ can be defined as,

$$f(t) = 11 \sin \pi t, \quad 0 < t < 2$$

Fourier transform of the function $f(t)$ is,

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Substitute the function $f(t) = 11 \sin \pi t$ in the equation.

$$F(\omega) = \int_0^2 11 \sin \pi t e^{-j\omega t} dt \dots\dots (1)$$

Step 3 of 5

From Euler's identity,

$$\sin \pi t = \frac{e^{j\pi t} - e^{-j\pi t}}{j2}$$

Substitute the expression for $\sin \pi t$ in equation (1).

$$\begin{aligned} F(\omega) &= 11 \int_0^2 \left(\frac{e^{j\pi t} - e^{-j\pi t}}{j2} \right) e^{-j\omega t} dt \\ &= \frac{11}{j2} \left[\int_0^2 e^{-j(\omega-\pi)t} dt - \int_0^2 e^{-j(\omega+\pi)t} dt \right] \\ &= -5.5j \left\{ \left[\frac{e^{-j(\omega-\pi)t}}{-j(\omega-\pi)} \right]_0^2 - \left[\frac{e^{-j(\omega+\pi)t}}{-j(\omega+\pi)} \right]_0^2 \right\} \end{aligned}$$

Step 4 of 5

Simplify the equation.

$$\begin{aligned} F(\omega) &= -5.5j \left\{ \left[\frac{e^{-j(\omega-\pi)2} - e^0}{-j(\omega-\pi)} \right] - \left[\frac{e^{-j(\omega+\pi)2} - e^0}{-j(\omega+\pi)} \right] \right\} \\ &= -5.5j \left\{ \frac{e^{-j2(\omega-\pi)} - 1}{-j(\omega-\pi)} - \frac{e^{-j2(\omega+\pi)} - 1}{-j(\omega+\pi)} \right\} \\ &= \frac{-5.5j}{-j} \left\{ \frac{e^{-j2(\omega-\pi)} - 1}{(\omega-\pi)} - \frac{e^{-j2(\omega+\pi)} - 1}{(\omega+\pi)} \right\} \\ &= 5.5 \frac{(e^{-j2(\omega-\pi)} - 1)(\omega+\pi) - (e^{-j2(\omega+\pi)} - 1)(\omega-\pi)}{(\omega-\pi)(\omega+\pi)} \\ &= 5.5 \frac{(\omega e^{-j2(\omega-\pi)} - \omega + \pi e^{-j2(\omega-\pi)} - \pi) - (\omega e^{-j2(\omega+\pi)} - \omega - \pi e^{-j2(\omega+\pi)} + \pi)}{\omega^2 - \pi^2} \\ &= 5.5 \frac{\omega(e^{-j2\omega} e^{j2\pi} - e^{-j2\omega} e^{-j2\pi}) + \pi(e^{-j2\omega} e^{j2\pi} + e^{-j2\omega} e^{-j2\pi}) - 2\pi}{\omega^2 - \pi^2} \end{aligned}$$

Step 5 of 5

It is known that, $e^{-j2\pi} = e^{j2\pi} = 1$

Substitute 1 for $e^{-j2\pi}$ and 1 for $e^{j2\pi}$ in equation of $F(\omega)$.

$$\begin{aligned} F(\omega) &= 5.5 \frac{\omega(e^{-j2\omega} - e^{-j2\omega}) + \pi(e^{-j2\omega} + e^{-j2\omega}) - 2\pi}{\omega^2 - \pi^2} \\ &= 5.5 \left(\frac{0 + 2\pi e^{-j2\omega} - 2\pi}{\omega^2 - \pi^2} \right) \\ &= \frac{11\pi}{\omega^2 - \pi^2} (e^{-j2\omega} - 1) \end{aligned}$$

Therefore, the Fourier transform of the sine-wave pulse is $F(\omega) = \frac{11\pi}{\omega^2 - \pi^2} (e^{-j2\omega} - 1)$.